

Q1.1 We can use the following property (#11) to do this question

$$\mathcal{Z}[k \ x(k)] = -z \frac{d}{dz} X(z)$$

Note that

$$\mathcal{Z}[1] = \frac{1}{1 - z^{-1}}$$

$$\left(\frac{z}{z-1}\right)' = \frac{(z-1) - z}{(z-1)^2} = -\frac{1}{(z-1)^2}$$

$$z\{k\} = -z\left(\frac{z}{z-1}\right)' = \frac{z}{(z-1)^2}$$

$$\left(\frac{z}{(z-1)^2}\right)' = \frac{(z-1)^2 - 2z(z-1)}{(z-1)^4} = -\frac{z+1}{(z-1)^3}$$

$$z\{k^2\} = -z\left(\frac{z}{(z-1)^2}\right)' = \frac{z(z+1)}{(z-1)^3} = \frac{z^2+z}{(z-1)^3}$$

$$\left(\frac{z^2+z}{(z-1)^3}\right)' = \frac{(2z+1)(z-1)^3 - 3(z-1)^2(z^2+z)}{(z-1)^6}$$

$$= \frac{(2z+1)(z-1) - 3(z^2+z)}{(z-1)^4}$$

$$= -\frac{z^2+4z+1}{(z-1)^4}$$

$$z\{k^3\} = -z\left(-\frac{z^2+4z+1}{(z-1)^4}\right) = \frac{z^3+4z^2+z}{(z-1)^4}$$

Q1.2

We can use the Complex Translation property of $\mathcal{Z}$ transform

$$\mathcal{Z} [x(t)e^{-at}] = X(ze^{aT})$$

as follows:

$$\begin{aligned}\mathcal{Z} [t^2] &= \frac{T^2 z^{-1}(1 + z^{-1})}{(1 - z^{-1})^3} \\ \mathcal{Z} [t^2 e^{-at}] &= \frac{T^2 (ze^{aT})^{-1} (1 + (ze^{aT})^{-1})}{(1 - (ze^{aT})^{-1})^3} \\ &= \frac{T^2 z^{-1} e^{-aT} (1 + z^{-1} e^{-aT})}{(1 - z^{-1} e^{-aT})^3}\end{aligned}$$

We can also use #21 (Table 2-2) property of $\mathcal{Z}$ transform:

$$\frac{\partial}{\partial a}x(t, a) = \frac{\partial}{\partial a}X(z, a)$$

Namely,

$$\mathcal{Z} [e^{-at}] = \sum_{k=0}^{\infty} e^{-akT} z^{-k} = \frac{1}{1 - z^{-1}e^{-aT}}$$

and

$$\mathcal{Z} [-te^{-at}] = \mathcal{Z} \left[\frac{\partial}{\partial a} e^{-at} \right] = \frac{\partial}{\partial a} \mathcal{Z} [e^{-at}] = \frac{\partial}{\partial a} \frac{1}{1 - z^{-1}e^{-aT}}$$

Repeating the above one more time, we have

$$\mathcal{Z} [t^2 e^{-at}] = \mathcal{Z} \left[\frac{\partial^2}{\partial a^2} e^{-at} \right] = \frac{\partial^2}{\partial a^2} \mathcal{Z} [e^{-at}] = \frac{\partial^2}{\partial a^2} \frac{1}{1 - z^{-1}e^{-aT}}$$

Evaluating the above differentiations, we have

$$\mathcal{Z} [t^2 e^{-at}] = \frac{T^2 z^{-1} e^{-aT} (1 + z^{-1} e^{-aT})}{(1 - z^{-1} e^{-aT})^3}$$

Q1.3

$$\begin{aligned}
 x(k) &= 9k2^{k-1} - 2^k + 3, \quad k \geq 0 \\
 &= \frac{9}{2}k2^k - 2^k + 3
 \end{aligned}$$

Hence

$$\begin{aligned}
 X(z) &= \mathcal{Z}[x(k)] = \sum_{k=0}^{\infty} x(k)z^{-k} \\
 &= \frac{9}{2} \sum_{k=0}^{\infty} k2^k z^{-k} - \sum_{k=0}^{\infty} 2^k z^{-k} + 3 \sum_{k=0}^{\infty} z^{-k} \\
 &= \frac{9}{2} \sum_{k=0}^{\infty} k(2z^{-1})^k - \sum_{k=0}^{\infty} (2z^{-1})^k + 3 \sum_{k=0}^{\infty} (z^{-1})^k
 \end{aligned}$$

Noting that

$$\mathcal{Z}[k] = \frac{z^{-1}}{(1 - z^{-1})^2}, \quad \mathcal{Z}[1] = \frac{1}{(1 - z^{-1})}$$

we have

$$\begin{aligned}
 X(z) &= \frac{9}{2} \frac{z^{-1}}{(1 - z^{-1})^2} \Big|_{z^{-1} \rightarrow 2z^{-1}} - \frac{1}{1 - z^{-1}} \Big|_{z^{-1} \rightarrow 2z^{-1}} + \frac{3}{1 - z^{-1}} \\
 &= \frac{2 + z^{-2}}{(1 - 2z^{-1})^2(1 - z^{-1})}
 \end{aligned}$$

Q1.4 From Lecture notes

$$\mathcal{Z} \left[\sum_{h=0}^k x(h) \right] = \frac{1}{1 - z^{-1}} \mathcal{Z} [x(k)]$$

Hence, for

$$y(k) = \sum_{h=0}^k a^h$$

we have

$$\mathcal{Z} [y(k)] = \frac{1}{1 - z^{-1}} \mathcal{Z} [a^k] = \frac{1}{(1 - z^{-1})(1 - az^{-1})}$$

Q1.5 With $T = 1$, we have

$$x(0) = x(1) = x(2) = 0, \quad x(3) = \frac{1}{3}, \quad x(4) = \frac{2}{3},$$

and

$$x(k) = 1 \text{ for all } k \geq 5$$

Hence

$$\begin{aligned} \mathcal{Z}[x(k)] &= x(0) + x(1)z^{-1} + x(2)z^{-2} + x(3)z^{-3} + x(4)z^{-4} \\ &\quad + x(5)z^{-5} + x(6)z^{-6} + \dots \\ &= 0 + 0z^{-1} + 0z^{-2} + \frac{1}{3}z^{-3} + \frac{2}{3}z^{-4} + z^{-5} + z^{-6} + \dots \\ &= \frac{1}{3}z^{-3} + \frac{2}{3}z^{-4} + z^{-5} [1 + z^{-1} + z^{-2} + z^{-3} + \dots] \\ &= \frac{1}{3}z^{-3} + \frac{2}{3}z^{-4} + \frac{z^{-5}}{1 - z^{-1}} \\ &= \frac{z^{-3}(1 + z^{-1} + z^{-2})}{3(1 - z^{-1})} \end{aligned}$$

Q2.1

$$X(z) = \frac{1 + 2z + 3z^2 + 4z^3 + 5z^4}{z^4} = 5 + 4z^{-1} + 3z^{-2} + 2z^{-3} + z^{-4}$$

Hence

$$x(0) = 5, \quad x(1) = 4, \quad x(2) = 3, \quad x(3) = 2, \quad x(4) = 1,$$

and

$$x(k) = 0 \quad \text{for } k \geq 5$$

Q2.2

$$X(z) = \frac{z^{-1}(0.5 - z^{-1})}{(1 - 0.5z^{-1})(1 - 0.8z^{-1})} = \frac{z(0.5z - 1)}{(z - 0.5)(z - 0.8)^2}$$

$$\frac{X(z)}{z} = \frac{(0.5z - 1)}{(z - 0.5)(z - 0.8)^2} = \frac{-25/3}{z - 0.5} + \frac{25/3}{z - 0.8} + \frac{-2}{(z - 0.8)^2}$$

$$X(z) = \frac{-25/3}{1 - 0.5z^{-1}} + \frac{25/3}{1 - 0.8z^{-1}} + \frac{-2z^{-1}}{(1 - 0.8z^{-1})^2}$$

Hence

$$x(k) = -\frac{25}{3}(0.5)^k + \frac{25}{3}(0.8)^k - 2k(0.8)^{k-1}, \quad k = 0, 1, 2, 3, \dots$$

Q2.3

$$\begin{aligned}x(0) &= \lim_{z \rightarrow \infty} X(z) = \lim_{z \rightarrow \infty} \frac{z^{-1}}{(1 - z^{-1})(1 + 1.3z^{-1} + 0.4z^{-2})} \\&= \frac{0}{(1 - 0)(1 + 0)} = 0\end{aligned}$$

Before applying the Final Value Theorem, need to check the poles of $X(z)$:

$$z = 1, \quad z = -0.5, \quad z = -0.8$$

As there is one pole at $z = 1$ and all the other poles have absolute values of less than 1, hence we can apply the Final Value Theorem.

$$x(\infty) = \lim_{z \rightarrow 1} (1 - z^{-1})X(z) = \lim_{z \rightarrow 1} \frac{z^{-1}}{1 + 1.3z^{-1} + 0.4z^{-2}} = \frac{1}{2.7}$$

Indeed,

$$\begin{aligned}X(z) &= \frac{z^{-1}}{(1 - z^{-1})(1 + 1.3z^{-1} + 0.4z^{-2})} \\&= \frac{1}{2.7} \left[\frac{1}{1 - z^{-1}} + \frac{3}{1 + 0.5z^{-1}} + \frac{-4}{1 + 0.8z^{-1}} \right]\end{aligned}$$

Hence

$$x(k) = \frac{1}{2.7} [1 + 3(-0.5)^k - 4(-0.8)^k], \quad k \geq 0$$

Obviously

$$\begin{aligned}x(0) &= \frac{1}{2.7}(1 + 3 - 4) = 0 \\x(\infty) &= \frac{1}{2.7}(1 + 0 + 0) = \frac{1}{2.7}\end{aligned}$$

Q2.4 Inversion Integral Method

$$X(z) = \frac{1 + z^{-1} - z^{-2}}{1 - z^{-1}} = \frac{z^2 + z - 1}{z(z - 1)}$$

$$X(z)z^{k-1} = \frac{(z^2 + z - 1)z^{k-1}}{z(z - 1)}$$

For $k = 0$,

$$X(z)z^{k-1} = \frac{z^2 + z - 1}{z^2(z - 1)}$$

and

$$\begin{aligned} x(0) &= \left[\text{Residual of } \frac{z^2 + z - 1}{z^2(z - 1)} \text{ at the simple pole } z = 1 \right] \\ &\quad + \left[\text{Residual of } \frac{z^2 + z - 1}{z^2(z - 1)} \text{ at the double pole } z = 0 \right] \\ &= \lim_{z \rightarrow 1} \left[(z - 1) \times \frac{z^2 + z - 1}{z^2(z - 1)} \right] \\ &\quad + \frac{1}{(2 - 1)!} \lim_{z \rightarrow 0} \frac{d^{2-1}}{dz^{2-1}} \left[z^2 \times \frac{z^2 + z - 1}{z^2(z - 1)} \right] \\ &= 1 + \lim_{z \rightarrow 0} \frac{(2z + 1)(z - 1) - (z^2 + z - 1)}{(z - 1)^2} \\ &= 1 + 0 = 1 \end{aligned}$$

For $k = 1$,

$$X(z)z^{k-1} = \frac{z^2 + z - 1}{z(z-1)}$$

and

$$\begin{aligned} x(1) &= \left[\text{Residual of } \frac{z^2 + z - 1}{z(z-1)} \text{ at the simple pole } z = 1 \right] \\ &\quad + \left[\text{Residual of } \frac{z^2 + z - 1}{z(z-1)} \text{ at the simple pole } z = 0 \right] \\ &= \lim_{z \rightarrow 1} \left[(z-1) \times \frac{z^2 + z - 1}{z(z-1)} \right] + \lim_{z \rightarrow 0} \left[z \times \frac{z^2 + z - 1}{z(z-1)} \right] \\ &= 1 + 1 = 2 \end{aligned}$$

For $k \geq 2$,

$$X(z)z^{k-1} = \frac{(z^2 + z - 1)z^{k-2}}{(z-1)}$$

Note: Only one pole at $z = 1$. No more pole at $z = 0$!

$$\begin{aligned} x(k) &= \left[\text{Residual of } \frac{(z^2 + z - 1)z^{k-2}}{(z-1)} \text{ at the simple pole } z = 1 \right] \\ &= \lim_{z \rightarrow 1} \left[(z-1) \times \frac{(z^2 + z - 1)z^{k-2}}{(z-1)} \right] = 1 \end{aligned}$$

Hence,

$$x(0) = 1, \quad x(1) = 2, \quad x(k) = 1 \text{ for } k \geq 2$$

Note: We can also use the Final Value Theorem to verify

$$x(\infty) = \lim_{z \rightarrow 1} (1 - z^{-1})X(z) = 1$$

Of course, given the choice, we would rather do the following:

$$\begin{aligned} X(z) &= \frac{1 + z^{-1} - z^{-2}}{1 - z^{-1}} = \frac{1 - z^{-1} + 2z^{-1} - 2z^{-2} + z^{-2}}{1 - z^{-1}} \\ &= 1 + 2z^{-1} + \frac{z^{-2}}{1 - z^{-1}} \end{aligned}$$

Hence

$$x(k) = \delta_0(k) + 2\delta_0(k-1) + \underline{1}(k-2)$$

Q2.5 (B-2-12)

$$\begin{aligned}X(z) &= \frac{z^{-3}}{(1 - z^{-1})(1 - 0.2z^{-1})} \\&= \frac{1}{z(z - 1)(z - 0.2)} \\&= \frac{5}{z} + \frac{1.25}{z - 1} + \frac{-6.25}{z - 0.2} \\&= 5z^{-1} + 1.25 \times \frac{z^{-1}}{1 - z^{-1}} - 6.25 \times \frac{z^{-1}}{1 - 0.2z^{-1}} \\&= z^{-1} \left[5 + \frac{1.25}{1 - z^{-1}} - \frac{6.25}{1 - 0.2z^{-1}} \right]\end{aligned}$$

Hence

$$\begin{aligned}x(k) &= \left[5\delta_0(k) + 1.25u(k) - 6.25(0.2)^k u(k) \right] \Big|_{k \rightarrow k-1} \\&= 5\delta_0(k - 1) + 1.25u(k - 1) - 6.25(0.2)^{k-1} u(k - 1)\end{aligned}$$

i.e.,

$$\begin{aligned}x(k) &= 0 \quad \text{for } k = 0, 1, 2 \\x(k) &= 1.25(1 - 0.2^{k-2}) \quad \text{for } k = 3, 4, 5, \dots\end{aligned}$$

However, the following will not work:

$$X(z) = \frac{z^{-3}}{(1 - z^{-1})(1 - 0.2z^{-1})} = \frac{A}{1 - z^{-1}} + \frac{B}{1 - 0.2z^{-1}}$$

as $X(z)$ above is not strictly proper in z^{-1} .

Instead, we must write

$$X(z) = \frac{z^{-2} z^{-1}}{(1 - z^{-1})(1 - 0.2z^{-1})} = \frac{Az^{-2}}{1 - z^{-1}} + \frac{Bz^{-2}}{1 - 0.2z^{-1}}$$

$$\Rightarrow z^{-3} = Az^{-2}(1 - 0.2z^{-1}) + Bz^{-2}(1 - z^{-1})$$

$$\Rightarrow A = 1.25, \quad B = -1.25$$

Hence,

$$x(k) = \left[1.25 - 1.25(0.2)^k \right] \Big|_{k \rightarrow k-2} = 1.25 \left[1 - (0.2)^{k-2} \right] u(k-2)$$

Q3.1.

Firstly, we determine the initial values of $x(k)$.

When $k=2$, we have

$$x(0) - 2x(-1) + x(-2) = f(-2) = 0$$

$$\Rightarrow x(0) = 0$$

When $k=-1$,

$$x(1) - 2x(0) + x(-1) = f(-1) = 0$$

$$\Rightarrow x(1) = 0$$

Then taking z -transform gives

$$\begin{aligned} z^2 X(z) - z^2 x(0) - z x(1) \\ - 2[zX(z) - z x(0)] + X(z) = 1 \end{aligned}$$

$$z^2 x(z) - 2z x(z) + x(z) = 1$$

$$(z^2 - 2z + 1) x(z) = 1$$

$$x(z) = \frac{1}{(z-1)^2}$$

$$= \frac{z^{-2}}{(1-z^{-1})^2}$$

$$= \frac{(1-z^{-1})^2 - (1-z^{-1}) + z^{-1}}{(1-z^{-1})^2}$$

$$= 1 - \frac{1}{(1-z^{-1})} + \frac{z^{-1}}{(1-z^{-1})^2}$$

$$\therefore x(k) = \delta(k) - 1(k) + k,$$

$$k=0, 1, 2, \dots$$

Q3.2

- (1) Use the z-Transform Table after doing partial fraction expansion

(2) Method Based on Inverse Laplace Transform of $X(s)$.

$$\begin{aligned} X(s) &= \frac{s+3}{(s+1)(s+2)} \\ &= \frac{2}{(s+1)} - \frac{1}{(s+2)} \end{aligned}$$

The inverse Laplace transform of this equation gives

$$x(t) = 2e^{-t} - e^{-2t}$$

Hence

$$\begin{aligned} X(z) &= \frac{2}{(1 - e^{-T}z^{-1})} - \frac{1}{(1 - e^{-2T}z^{-1})} \\ &= \frac{1 + e^{-T}(1 - 2e^{-T})z^{-1}}{(1 - e^{-T}z^{-1})(1 - 2e^{-2T}z^{-1})} \end{aligned}$$

3.3

From Figure 3.2 we obtain

$$\begin{aligned}C(s) &= G(s)E^*(s) \\ E(s) &= R(s) - H_2(s)M^*(s) \\ M(s) &= H_1(s)G(s)E^*(s)\end{aligned}$$

By taking the starred Laplace transforms of the preceding equations, we have

$$\begin{aligned}C^*(s) &= G^*(s)E^*(s) \\ E^*(s) &= R^*(s) - H_2^*(s)M^*(s) \\ M^*(s) &= [GH_1(s)]^*E^*(s)\end{aligned}$$

Hence

$$E^*(s) = R^*(s) - H_2^*(s) [GH_1(s)]^* E^*(s)$$

or

$$E^*(s) = \frac{R^*(s)}{1 + H_2^*(s)[GH_1(s)]^*}$$

$$C^*(s) = \frac{G^*(s)R^*(s)}{1 + H_2^*(s)[GH_1(s)]^*}$$

and

$$\frac{C^*(s)}{R^*(s)} = \frac{G^*(s)}{1 + H_2^*(s)[GH_1(s)]^*}$$

$$\frac{C(z)}{R(z)} = \frac{G(z)}{1 + H_2(z)[GH_1(z)]}$$

~~Q4~~.1 (B-3-17)

$$C(s) = G_2(s)M^*(s) \quad (1)$$

$$M(s) = G_1(s)E^*(s) - C(s) \quad (2)$$

$$E(s) = R(s) - H(s)C(s) \quad (3)$$

Substituting (1) into (2) and (3) yields

$$M(s) = G_1(s)E^*(s) - G_2(s)M^*(s) \quad (4)$$

$$E(s) = R(s) - H(s)G_2(s)M^*(s) \quad (5)$$

Now apply the * transformation on (4) and (5)

$$M^*(s) = G_1^*(s)E^*(s) - G_2^*(s)M^*(s) \quad (6)$$

$$E^*(s) = R^*(s) - (HG_2)^*(s)M^*(s) \quad (7)$$

Substituting (7) into (6) gives

$$M^*(s) = G_1^*(s)[R^*(s) - (HG_2)^*(s)M^*(s)] - G_2^*(s)M^*(s)$$

hence

$$M^*(s) = \frac{G_1^*(s)R^*(s)}{1 + G_1^*(s)(HG_2)^*(s) + G_2^*(s)}$$

Also

$$C(s) = G_2(s)M^*(s) = \frac{G_2(s)G_1^*(s)R^*(s)}{1 + G_1^*(s)(HG_2)^*(s) + G_2^*(s)}$$

and

$$C^*(s) = \frac{G_2^*(s)G_1^*(s)R^*(s)}{1 + G_1^*(s)(HG_2)^*(s) + G_2^*(s)}$$

$$C(z) = \frac{G_2(z)G_1(z)R(z)}{1 + G_1(z)HG_2(z) + G_2(z)}$$

Q4.2 (B-3-18)

$$C(s) = \left(\frac{K}{s} \frac{1 - e^{-Ts}}{s} \right) E^*(s)$$

$$E(s) = R(s) - C(s) = R(s) - \left(\frac{K}{s} \frac{1 - e^{-Ts}}{s} \right) E^*(s)$$

Thus

$$E^*(s) = R^*(s) - C^*(s) = R^*(s) - \left(\frac{K}{s} \frac{1 - e^{-Ts}}{s} \right)^* E^*(s)$$

and

$$E^*(s) = \frac{R^*(s)}{1 + \left(\frac{K}{s} \frac{1 - e^{-Ts}}{s} \right)^*}$$

Hence

$$C(s) = \frac{\frac{K}{s} \frac{1 - e^{-Ts}}{s}}{1 + \left(\frac{K}{s} \frac{1 - e^{-Ts}}{s} \right)^*} R^*(s)$$

and

$$C^*(s) = \frac{\left(\frac{K}{s} \frac{1 - e^{-Ts}}{s} \right)^*}{1 + \left(\frac{K}{s} \frac{1 - e^{-Ts}}{s} \right)^*} R^*(s)$$

$$C(z) = \frac{\mathcal{Z} \left[\frac{K}{s} \frac{1 - e^{-Ts}}{s} \right]}{1 + \mathcal{Z} \left[\frac{K}{s} \frac{1 - e^{-Ts}}{s} \right]} R(z)$$

Since $R(z) = \frac{1}{1 - z^{-1}}$ and

$$\begin{aligned}\mathcal{Z} \left[\frac{1 - e^{-Ts}}{s} \frac{K}{s} \right] &= (1 - z^{-1}) \mathcal{Z} \left[\frac{K}{s^2} \right] \\ &= (1 - z^{-1}) \times \frac{KTz^{-1}}{(1 - z^{-1})^2} \\ &= \frac{Kz^{-1}}{1 - z^{-1}}\end{aligned}$$

where $T = 1s$ is used, we have

$$C(z) = \frac{Kz^{-1}}{1 + (K - 1)z^{-1}} \frac{1}{1 - z^{-1}}$$

If $K \neq 0$, we have

$$C(z) = \frac{-1}{1 + (K - 1)z^{-1}} + \frac{1}{1 - z^{-1}}$$

and

$$c(k) = 1 - (1 - K)^k, \quad k = 0, 1, 2, \dots$$

Note that $c(k)$ is convergent if $|1 - K| < 1$, i.e., if $0 < K < 2$.

If $K > 2$ or $K < 0$, $c(k) \rightarrow \infty$ as $k \rightarrow \infty$ (i.e., divergent).

If $K = 2$, $c(k)$ oscillates between 0 and 2, and will not converge to a single value.

If $K = 0$, we have $C(z) = 0$, hence, $c(k) = 0$ for all $k \geq 0$

(i.e., convergent).

4.3

From Figure 4.3, we obtain

$$\begin{aligned}M(z) &= \frac{K_I}{1 - z^{-1}} [R(z) - C(z)] \\&\quad - [K_p + K_D(1 - z^{-1})]C(z) \\&= \frac{K_I}{1 - z^{-1}} R(z) \\&\quad - \left[\frac{K_I}{1 - z^{-1}} + K_p + K_D(1 - z^{-1}) \right] C(z)\end{aligned}$$

$$\begin{aligned}C(z) &= G(z)M(z) \\&= G(z) \frac{K_I}{1 - z^{-1}} R(z) \\&\quad - G(z) \left[\frac{K_I}{1 - z^{-1}} + K_p + K_D(1 - z^{-1}) \right] C(z)\end{aligned}$$

4.5

$$\{1 + G(z) \left[\frac{K_I}{1-z^{-1}} + K_p + K_D(1-z^{-1}) \right] \} C(z)$$

$$= G(z) \frac{K_I}{1-z^{-1}} R(z)$$

$$\frac{C(z)}{R(z)} = \frac{G(z) \frac{K_I}{1-z^{-1}}}{1 + G(z) \left[\frac{K_I}{1-z^{-1}} + K_p + K_D(1-z^{-1}) \right]}$$